

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Why do you need data structure? Data Structure is an abstract data type. Do you agree with this Statement? If so, explain with example. 7
- b) Explain Divide and Conquer with an example. 8

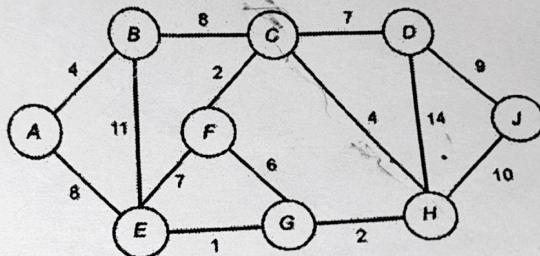
OR

What is Big OH notation in algorithm analysis? Calculate upper bound time complexity of following snippets of code.

```
int j,k;  
for (j=0; j < N; j=j*2)  
    sum += j;
```

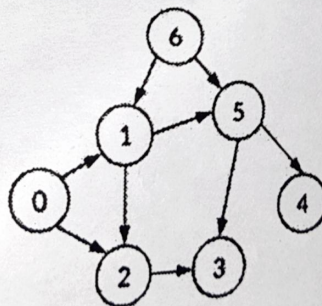
2. a) Differentiate between recursion and iteration. Write recursive algorithm for solving Tower of Hanoi problem. 7
- b) Convert following infix expression to postfix using stack. 8
- $A + B * (C \wedge D - E) \wedge (F + G * H) - I$
3. a) How does the circular queue overcome the problem of linear queue? Explain with algorithm how insertion and deletion are performed in circular queue using array. 7
- b) What is the advantage of doubly linked list over singly linked list? Write algorithms to insert and delete a node at the beginning of singly linked list. 8
- a) Differentiate array with linked list. Explain the linked implementation of queue. 7
- b) Construct a binary search tree after inserting the following values. 12, 5, 15, 13, 17, 3, 7, 1, 9. Show your binary tree after deletion of 7, 15. 8

5. a) What is balance factor in AVL tree? Construct an AVL tree from following sequence of number inserted in an order of empty binary tree: 20, 24, 29, 8, 3, 13, 27, 17, 14, 9, 1. 8
- b) Explain how do you use divide and conquer strategy to sort the following list using quick sort: 7, 9, 45, 12, 56, 90, 3, 8, 50. Also write the best, average and worst case time complexity of quick sort for simple partition case. 7
6. a) What is the principle of hash function in searching? For the data {1,16,49,25,64,0,81,4,9} and a hash function $h(x) = x \bmod 10$, find the resulting: 8
- Hash Table using linear and quadratic probing.
 - Hash table using chaining.
- b) What is transitive closure of a graph. Illustrate with simple example. Find the minimum spanning tree using Kruskal algorithm of following graph: 7



OR

Define topological sort? Perform the topological sort for the following graph:



2×5

7. Write short notes on: (Any two)

- Priority queue
- Traversing in a binary tree
- Bubble Sort